

### A) Precision of Estimates – Standard Errors and Confidence Intervals

We noted earlier (the last paragraph of sub-section 7.5.1) that the sample mean (an estimate of the population mean ' $\mu$ ') will have different values in repeated samples drawn from the population and none of these may be equal to ' $\mu$ '. Suppose that the repeated samples drawn from the population are *random* samples. The sample mean computed from a *random sample* is a *random variable*. So is the sampling error, that is, the difference between ' $\mu$ ' and the sample mean. The values of the sample means (and the corresponding errors in the estimate of ' $\mu$ ') computed from the repeated random samples drawn from the population are the values assumed by this random variable with probabilities associated with drawing the corresponding samples. These will trace out a frequency distribution that will approach a probability distribution when the *number* of random samples drawn increases indefinitely. *The probability distribution of sample means computed from all possible random samples from the population is called the **sampling distribution of the sample mean***. The sampling distribution of the sample mean has a mean and a standard deviation. *The sample mean is said to be an **unbiased estimator** of the population mean if the mean of the sampling distribution of the sample mean is equal to the mean of the parent population, say,  $\mu$* . In general, an estimator " $t$ " of a population parameter " $\theta$ " is an unbiased estimator of " $\theta$ " if the mean of the sampling distribution of " $t$ ", or the expected value of the random variable " $t$ ", is equal to " $\theta$ ". In other words, the mean of the estimates of the parameter made from all possible samples drawn from the population will be equal to the value of the parameter. Otherwise, it is said to be a *biased estimate*. Supposing the mean of the sampling distribution of sample mean is  $K\mu$  or  $K+\mu$ , where  $K$  is a constant. The bias in the estimate can be easily corrected in such cases by adopting  $m/K$  or  $(m - K)$  as the estimator of the population mean.

The variance of the sampling distribution of the sample mean is called the ***sampling variance of the sample mean***. The standard deviation of the sampling distribution of sample means is called *the **standard error (SE) of the sample mean***. It is also called the ***standard error of the estimator (of the population mean)***, as the sample mean is an estimator of the population mean. *The standard error of the sample mean is a measure of the variability of the sample mean about the population mean or a measure of the **precision** of the sample mean as an estimator of the population mean*. The ratio of the standard deviation of the sampling distribution of sample means and the mean of the sampling distribution is called the ***coefficient of variation (CV)*** of the sample mean or the ***relative standard error (RSE)*** of the sample mean. That is,

$$CV \text{ or } RSE = C = \text{standard deviation} / \text{mean} \quad (2.14)$$

CV (or RSE) is a free number or is dimension-less, while the mean and the standard deviation are in the same units as the variable ' $y$ '. (These definitions can easily be generalised to the sampling distribution of any sample statistic and its SE and RSE.)

We have talked about the unbiasedness and precision of the estimate made from the sample. What more can we say about the precision of the estimate and other characteristics of the estimate? *This is possible if we know the nature of the sampling distribution of the estimate.*

The nature of the sampling distribution of, say, the sample mean, or for that matter any statistic, depends on the nature of the population from which the random sample is drawn. If the parent population has a normal distribution with mean  $\mu$  and variance  $\sigma^2$  or, in short notation,  $N(\mu, \sigma^2)$ , the sampling distribution of the sample mean, based on a random sample drawn from this, is  $N(\mu, \sigma^2/n)$ . In other words, the variability of the sample mean is much smaller than that of the variable of the population and it also *decreases* as the sample size *increases*. ***Thus, the precision of the sample mean as an estimate of the population mean increases as the sample size increases.***

As we know, the normal distribution  $N(\mu, \sigma^2)$  has the following properties:

- i) Approximately 68% of all the values in a normally distributed population lie within a distance of one standard deviation (plus and minus) from the mean,
- ii) Approximately 95% of all the values in a normally distributed population lie within a distance of 1.96 standard deviation (plus and minus) of the mean,
- iii) Approximately 99% of all the values in a normally distributed population lie within a distance of 2.576 standard deviation (plus and minus) of the mean.

The statement at (iii) above, for instance, is equivalent to saying that the population mean  $\mu$  will lie between the observed values  $(\bar{y} - 2.576 \sigma)$  and  $(\bar{y} + 2.576 \sigma)$  in 99% of the random samples drawn from the population  $N(\mu, \sigma^2)$ . Applying this to the sampling distribution of the sample mean, which is  $N(\mu, \sigma^2/n)$ , we can say that

$$\text{Pr.}[(\bar{m} - 2.576 \sigma / \sqrt{n}) < \mu < (\bar{m} + 2.576 \sigma / \sqrt{n})] = 0.99 \quad (2.15)$$

or that the population mean  $\mu$  will lie between the limits computed from the sample, namely,  $(\bar{m} - 2.576 \sigma / \sqrt{n})$  and  $(\bar{m} + 2.576 \sigma / \sqrt{n})$  in 99% of the samples drawn from the population. This is an ***interval estimate***, or a ***confidence interval***, for the parameter with a ***confidence coefficient of 99%*** derived from the sample.

***The general rule for constructing a confidence interval of the population mean with a confidence coefficient of 99% is: the lower limit of the confidence interval is given by the “estimate of the population mean minus 2.576 times the standard error of the estimate” and the upper limit of the interval by the “estimate plus 2.576 times the standard error of the estimate”.*** (2.16)

#### B) Assessment of Precision – Unknown Population Variance

If the parent population is distributed as  $N(M, \sigma^2)$  and  $\sigma^2$  is not known, we make use of an estimate of  $\sigma^2$ . The statistic ‘ $s^2$ ’ given in formula 2.6 can be one such, but this is not an unbiased estimate of  $\sigma^2$  as  $E(s^2) = [(n-1)/n] \sigma^2$ . We, therefore, by using (2.6) and (2.7) have:

$$v(y) = [ns^2 / (n-1)] \text{ as an unbiased estimate of } \sigma^2. \quad (2.17)$$

$$v(y) = [1/(n-1)] [ss]^2 \text{ or, } v(y) = [(1/(n-1)) \sum y_i^2 - nm^2] \quad (2.18)$$

As the sampling variance of the sample mean 'm' is  $\sigma^2/n$ , an unbiased estimate  $v(m)$  of the sampling variance will be  $v(y)/n$ . Let us now consider the statistic defined by the ratio,

$$t = (m - M) / [\sqrt{v(y)} / \sqrt{n}] \quad (2.19)$$

The numerator is a random variable distributed as  $N(0, \sigma^2/n)$  and the denominator is the square root of the unbiased estimate of its variance. The sampling distribution of the statistic 't' is the Student's t-distribution with  $(n - 1)$  degrees of freedom. It is a symmetric distribution. A confidence interval can now be constructed for the population mean M from the selected random sample, say with a confidence coefficient of  $(1 - \alpha)\%$ . The values of ' $t_\alpha$ ' for different values of  $\alpha = \Pr.[t > t_\alpha] + \Pr.[(-t) < (-t_\alpha)] = 2 \Pr.[t > t_\alpha]$  and different degrees of freedom have been tabulated in, for instance, Rao, C.R. and Others (1966). The confidence interval with a confidence coefficient  $(1 - \alpha)$  for the population mean M would be as in 2.19 below – easily computed from the sample observations.

$$[m - t_\alpha \sqrt{v(m)} < M < m + t_\alpha \sqrt{v(m)}] \quad (2.20)$$

*We note that the rule 2.16 applies here also except that we use (i) the square root of the unbiased estimate of the sampling variance of the estimate of the population mean in the place of the standard error of the estimate of the population mean, and (ii) the relevant value of the 't' distribution instead of the normal distribution .....* (2.21)

We have so far dealt with parent populations that are normally distributed. What will be the nature of the sampling distribution of the sample mean when the **parent population is not normally distributed**? We examine this question in the next sub-section C.

### C) Assessment of Precision–Parent Population has a Non-Normal Distribution

The Central Limit Theorem ensures that, even if the population distribution is not normal,

- the sampling distribution of the sample mean will have a mean equal to the population mean *regardless* of the sample size, and
- as the sample size increases, the sampling distribution of the sample mean approaches the normal distribution.

Thus for large 'n' (sample size), say 30 or more, we can proceed with the steps mentioned in sub-section A above. Further, the Student's t-distribution also approaches the normal distribution as 'n' becomes large so that we can use the statistic 't' in sub-section B as a normally distributed variable with mean 0 and unit variance for samples of size 30 or more. We may then adopt the procedure outlined in sub-section A.

### D) Determination of Sample Size

Random sampling methods also help in determining the sample size that is required to attain a desired level of precision. This is possible because the standard error and the coefficient of variation C.V. of the estimate, say, sample mean 'm', are functions of 'n', the sample size. C.V. is usually very stable over the years and its value available from past data can be used for determining the

sample size. We can specify the value of C.V. of the sample mean that we desire as, say,  $C(m)$  and calculate the sample size with the help of prior knowledge of the population C.V., namely,  $C$ . That is,

$$C(m) = C / \sqrt{n} ; \text{ so that } \sqrt{n} = C/C(m), \text{ or, } n = [C/C(m)]^2 \quad (2.22)$$

Or we can define *the desired precision in terms of the error that we can tolerate in our estimate* of 'M' (**permissible error**) and link it with the desired value of  $C(m)$ . Then,

$$n = [2.576C / e]^2, \text{ where the permissible error } e = |(m - M)| / M. \quad (2.23)$$

**If the sanctioned budget is  $F$  for the survey:** Let the cost function be of the form  $F_0 + F_1n$ , consisting of two components – overhead cost and cost per unit to be surveyed. As this is fixed as  $F$ ,  $F = F_0 + F_1n$ , and the sample size becomes  $n = (F - F_0)/F_1$ . The coefficient of variation of  $C(m)$  is not at our choice in this situation since it gets fixed once 'n' is determined. We can, however, determine the error in the estimate of 'm' from this sample (*in terms of the RSE of m*), if the population CV,  $C$  is known. If further we suppose that the loss in terms of money is proportional to the value of RSE of  $m$ , say, Rs. ' $l$ ' per 1% of RSE of 'm', the total cost of the survey becomes,  $L(n) = F_0 + F_1n + l(C/\sqrt{n})$ . **We can then determine the sample size that minimises this new cost (which includes the cost arising out of loss).** Differentiating  $L(n)$  w.r.t  $n$  and equating to zero and simplifying,

$$n = [(l/2)(C/F_1)]^{2/3} \quad (2.24)$$

See also the sub-section below on stratified sampling.

#### 7.5.4 Methods of Random Sampling

We have so far dealt with random samples drawn from a population. We did not specify the size of the population. We had assumed that the population is infinite in size. In practice, a population may have a size  $N$ , however, large. Let us, therefore, consider drawing random samples of size 'n' from a population of size 'N'. We shall consider the following methods of random sampling:

- a) Simple Random Sampling (With Replacement) [SRSWR],
- b) Simple Random Sampling (Without Replacement) [SRSWOR]
- c) Interpenetrating Sub-Samples (I-PSS),
- d) Systematic Sampling (sys),
- e) Sampling with Probability Proportional to Size (pps)
- f) Stratified Sampling (sts),
- g) Cluster Sampling (cs) and
- h) Multi-Stage Sampling (mss)

We shall indicate in the following sections a description of the above methods, the relevant operational procedure for drawing a sample and the expressions/formulae for (a) the estimator of the population mean/total/ proportion, (b) the sampling variance of the sample mean/total/population and (c) unbiased estimate of the sampling variance.

- 1) Briefly describe the meaning of the terms ‘sampling distribution of a statistic’ and also what is meant by unbiased estimator of a parameter.

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- 2) How is random sampling procedure helpful in correcting for the bias of an estimate? Illustrate this with the help of an example.

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- 3) Explain the terms ‘coefficient of variation’ and ‘relative standard error’.

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#### 7.5.4.1 Simple Random Sampling with Replacement (SRSWR)

**The method:** This method of drawing samples at random ensures that (i) each item in the population has an equal chance of being included in the sample and (ii) each possible sample has an equal probability of getting selected. Let us select a sample of ‘n’ units from a population of ‘N’ units by simple random sampling with replacement (SRSWR). We select the first unit at random, note its identity particulars for collection of data and place it back in the population. We choose at random another unit – this could turn out to be the same unit selected earlier or a different one, note its identity particulars and place it back. We repeat this process ‘n’ times to get an SRSWR sample of size ‘n’. In such a sample one or more units *may* occur more than once. A sample of ‘n’ distinct units is also possible. It can be shown that the number of possible samples that can be selected by SRSWR method is  $N^n$  and that the probability of any one sample being chosen is  $1/N^n$ .

#### Operational procedure for selection of the sample by SRSWR method:

Tables of Random Numbers are used for drawing random samples. These tables contain a series of four-digit (or five-digit or ten-digit) random numbers. Supposing a sample of 30 units is to be selected out of a population of 3000 units. First allot one number from the set of numbers to 0001 to 3000 as the identification number to each one of the population units. The problem of drawing the sample of size 30 then reduces to that of selecting 30 random numbers, one after another, from the random number tables. Turn to a page of the Tables *at random* and start noting down, from the first left-most column of (four or five or ten-digit) random numbers, the first four digits of the numbers from the top of the column downwards. Continue this operation on to the second column till the required sample size of 30 is selected. If any of the random numbers that comes up is more than 3000, reject it. If some numbers ( $< 3000$ ) get repeated in the process, it means that the corresponding units of the population would be selected more than once, this being sampling with replacement.

### Estimators from SRSWR samples (using notations set down earlier):

$$m_{\text{srswr}} = (1/n) \sum_{i=1}^n y_i \quad (i = 1 \text{ to } n) \text{ is an unbiased estimator for } M \quad (2.25)$$

**Note:** If a unit gets selected in the sample more than once, the corresponding value of  $y_i$  will also have to be repeated as many times in the summation for calculating  $m_{\text{srswr}}$ .

$$\text{Sampling Variance of } m_{\text{srswr}} : V(m_{\text{srswr}}) = \sigma^2/n = [1/n] [E(y_i^2) - M^2] \quad (2.26)$$

$$\text{Standard Error of } m_{\text{srswr}} : SE(m_{\text{srswr}}) = \sigma / \sqrt{n} \quad (2.27)$$

$$\text{CV or RSE of } m_{\text{srswr}} : C(m_{\text{srswr}}) = (1/\sqrt{n})(\sigma / M) = C(y)/\sqrt{n}. \quad (2.28)$$

Note that the sampling variance, SE and CV(RSE) of the sample mean in SRSWR is much less than SE and CV of the variable  $y$  and these decrease as the sample size increases. The precision of the sample mean in SRSWR, as an estimator of  $M$  increases as the sample size increases. However, the extent of decrease in the standard error will not be commensurate with the size of the increase in the sample size. We would need an unbiased estimator of  $\sigma^2$ , as  $\sigma^2$  may not be known. This is

$$v(y) = [1/(n-1)][ss]^2 \quad (2.29);$$

$$\text{Therefore, } v(m_{\text{srswr}}) = v(y)/n \quad (2.30)$$

$$\text{an unbiased estimate of } Y, \text{ or, } Y_{\text{srswr}}^* = Nm \quad (2.31)$$

$$V(Y_{\text{srswr}}^*) = N^2 (\sigma^2 / n) \quad (2.32)$$

$$v(Y_{\text{srswr}}^*) = N^2 (1/n)[1/(n-1)] \sum_{i=1}^n (y_i - m_{\text{srswr}})^2 \quad (2.33)$$

$$\text{the sample proportion 'p}_{\text{srswr}}' \text{ is an unbiased estimate of } P \quad (2.34)$$

$$V(p_{\text{srswr}}) \text{ is } PQ/n \quad (2.35); \quad \text{and} \quad v(p_{\text{srswr}}) = npq/(n-1) \quad (2.36)$$

$$C(p_{\text{srswr}}) = \sqrt{[(1/n)(PQ)]/P} = [1/\sqrt{n}]\sqrt{[Q/P]}. \quad (2.37)$$

Confidence intervals for the population mean/proportion and the sample size for a given level of precision and/or permissible error can now be derived easily.

### 7.5.4.2 Simple Random Sampling without Replacement (SRSWOR)

**The Method:** This method of sampling is the same as SRSWR but for one difference. If a unit is selected, it is *not* placed back before the next one is selected. This means that no unit gets repeated in a sample. Operationally, we draw random numbers between 1 and  $N$  and if a random number comes up again, it is rejected and another random number is selected. This process is repeated till ' $n$ ' *distinct* units are selected. It can be shown that the number of samples of size  $n$  that may be selected from a population of ' $N$ ' units by this method is  ${}_N C_n = N! / [(N-n)! n!] = [N(N-1)(N-2) \dots (N-n+1)] / [n(n-1)(n-2) \dots 1]$ . The probability  $P_{\text{srswor}}(S)$  of any one of the samples being chosen is,  $1/{}_N C_n$ .

### Estimators from SRSWOR samples:

$$m_{\text{srswor}} = (1/n) \sum_{i=1}^n y_i, \quad i = 1 \text{ to } n, \text{ is an unbiased estimator of } M \quad (2.38)$$

$$\begin{aligned} V(m_{\text{srswor}}) &= [(N-n)/(N-1)] [\sigma^2/n] \\ &= [(N-n)/(N-1)] [1/n] [(1/N) \sum_{i=1}^N (Y_i - M)^2], \quad i = 1 \text{ to } N \end{aligned} \quad (2.39)$$

$$V(m)_{\text{srswor}} < V(m)_{\text{srswr}} \text{ since } (N - n) / (N - 1) \text{ is less than } 1 \text{ for } n > 1, \quad (2.40)$$

Both  $m_{\text{srswor}}$  and  $m_{\text{srswr}}$  are unbiased estimators of  $M$  but  $m_{\text{srswor}}$  is a more efficient estimator of  $M$  than  $m_{\text{srswr}}$ . The factor  $[(N - n)/(N - 1)]$  in (2.40) is called the **finite population correction or finite population multiplier**. The finite population correction required for finite population need not, however, be used when the sampling fraction  $(n / N)$  is less than 0.05.

$$\begin{aligned} v(m_{\text{srswor}}) &= [(N - n)/N][1/n][1/(n - 1)][\sum_{i=1}^n (y_i - m)^2], \quad i, i=1 \text{ to } n \\ &= [(N - n)/N][1/n][1/(n - 1)][ss]^2, \end{aligned} \quad (2.41)$$

$$\text{Unbiased estimate of population total } Y, \text{ that is } Y_{\text{srswor}}^* = Nm_{\text{srswor}} \quad (2.42)$$

$$V(Y_{\text{srswor}}^*) = N^2 V(m_{\text{srswor}}) \quad (2.43)$$

$$\text{unbiased estimate of } V(Y_{\text{srswor}}^*), \text{ namely, } v(Y_{\text{srswor}}^*) = N^2 v(m_{\text{srswor}}) \quad (2.44)$$

$$C(Y_{\text{srswor}}^*) = C(m_{\text{srswor}}) \quad (2.45)$$

the sample proportion 'p' is an unbiased estimate of 'P' in SRSWOR also. (2.46)

$$V(p) = [(N - n) / (N - 1)] [PQ/n], \text{ where } P + Q = 1 \quad (2.47)$$

$$v(p) = [(N - n)/N] [pq/(n - 1)], \text{ where } p + q = 1 \quad (2.48)$$

$$C(p) = \sqrt{[(N - n)/(N - 1)]} \sqrt{[(1/n)]} \sqrt{[Q/P]} \text{ and} \quad (2.49)$$

### Check Your Progress 3

- 1) A population has 80 units. The relevant variable has a population mean of 8.2 and a variance of 4.41. These SRSWR samples of size (i) 16, (ii) 25 and (iii) 49 are drawn from the population. What is the standard error (SE) of the sample means from the three samples? Is the extent of reduction in SE commensurate with that of the increase in sample size?

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- 2) What are the results when the sampling method in drawing the three samples in problem 1 above is changed to SRSWOR? What is your advice regarding the choice between increasing the sample size and changing the sampling method from SRSWR to SRSWOR?

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- 3) Indicate whether the following statements are true (T) or false (F). If false, what is the correct position?

- 1) The standard error of the sample mean decreases in direct proportion to the sample size. (T/F)

- 2) SRSWOR method of sampling is more advantageous than SRSWR for a sampling fraction of 0.02 (T/F)

- 3) If  $Y^* = Nm$  and Variance of  $m$  is  $V(m)$ , the variance of  $Y^*$  is  $NV(m)$ . (T/F)

### 7.5.4.3 Interpenetrating Sub-Samples (I-PSS)

Suppose a sample is selected in the form of two or more sub-samples drawn according to the same sampling method so that each such sub-sample provides a valid estimate of the population parameter. The sub-samples drawn in this way are called *interpenetrating sub-samples (I-PSS)*. This is operationally convenient, as the different sub-samples could be allotted to different investigators. The sub-samples need not be independently selected. There is, however, an important advantage in selecting *independent interpenetrating sub-samples*. *It is then possible to easily arrive at an unbiased estimate of the variance of the estimator even in cases where the sampling method/design is complex and the formula for the variance of the estimator is complicated.*

Let  $\{t_i\}$ ,  $i = 1, 2, \dots, h$  be unbiased estimates of a parameter  $\theta$  based on 'h' independent interpenetrating sub-samples. Then,

$$t = (1/h) \sum_{i=1}^h t_i, \quad (\sum_{i=1}^h t_i, i = 1 \text{ to } h) \text{ is an unbiased estimate of } \theta \quad (2.50)$$

$$v(t) = [1/h(h-1)] \sum_{i=1}^h (t_i - t)^2, \quad (\sum_{i=1}^h (t_i - t)^2, i = 1 \text{ to } h) \text{ is an unbiased estimate of } V(t) \quad (2.51)$$

If the unbiased estimator 't' of the parameter  $\theta$  is *symmetrically distributed* (for example, normally distributed), the probability of the parameter  $\theta$  lying between the maximum and the minimum of the 'h' estimates of  $\theta$  obtained from the 'h' sub-samples is given by:

$$\text{Prob.}[\text{Min of } \{t_1, t_2, \dots, t_h\} < \theta < \text{Max of } \{t_1, t_2, \dots, t_h\}] = [1 - (1/2)^{(h-1)}] \quad (2.52)$$

This is a confidence interval for  $\theta$  from the sample. The probability increases rapidly with the number of I-P sub-samples – from 0.5 (two sub-samples) to 0.875 (four sub-samples).

### 7.5.4.4 Systematic Sampling

**The Method:** Let  $\{U_i\}$ ,  $i = 1, 2, \dots, N$  be the units in a population. Let 'n' be the size of the sample to be selected. Let 'k' be the integer nearest to  $N/n$  – denoted usually as  $[N/n]$  — the reciprocal of the sampling fraction. Let us choose a random number from 1 to k, say, 'r'. We then choose the  $r^{\text{th}}$  unit, that is,  $U_r$ . Thereafter, we select every  $k^{\text{th}}$  unit. In other words, we select the units  $U_r, U_{r+k}, U_{r+2k}, \dots$ . This method of sampling is called *systematic sampling with a random start*. 'r' is known as *the random start* and 'k' *the sampling interval*. There would thus be 'k' possible systematic samples, each corresponding to one random start from 1 to k. The sample corresponding to the random start 'r' will be

$$\{U_{r+jk}\}, \quad j = 0, 1, 2, \dots, r+jk \leq N.$$

The sample size of all the 'k' systematic samples will be 'n' if  $N = nk$ . All the 'k' systematic samples will not have a sample size 'n' if  $N \neq nk$ . For example, if we have a sample of 100 units and we wish to select systematic samples of size 14, the sampling interval is  $k = [100/14]$  or 7. The samples with the random starting 1 and 2 will be of size 15 while the other 5 systematic samples (with random starts 3 to 7) will be of size 14.

In systematic sampling, units of a population could thus be selected at a uniform interval that is measured in time, order or space. We can for instance choose a sample of nails produced by a machine for five minutes at the interval

of every two hours to test whether the machine is turning out nails as per the desired specifications. Or, we could arrange the income tax returns relating to an area in the order of increasing gross income returned and select every fiftieth tax return for a detailed examination of the income of assesses of the area. Systematic samples are thus operationally easier to draw than SRSWR or SRSWOR samples. Only one random number needs to be chosen for selecting a systematic sample.

### Estimators from Systematic Samples:

**An unbiased estimator of the population mean  $M$**  based on a systematic sample is given by a **slight variant of the sample mean**, namely,

$$m_{sys}^* = (k/N) \sum_{i=1}^{n^*} y_i, \quad i = 1 \text{ to } n^*, \quad n^* \text{ is the size of the selected sample and } k \text{ the sampling interval} \quad (2.53)$$

If  $N = nk$ ,  $m_{sys}^* = m$  the sample mean. If  $N \neq nk$ , there is a bias in using the sample mean as the estimator for  $M$ , and

*the bias in using the sample mean as an estimator of  $M$  is likely to be small in the case of systematic samples selected from a large population.* (2.54)

The disadvantages, referred to above, in systematic sampling, namely,  $N$  not being a multiplier of the sample size  $n$  and the sample mean not being an unbiased estimator of the population mean can be overcome by adopting a procedure called **Circular Systematic Sampling (CSS)**. If ' $r$ ' is the random start, and  $k$  the integer nearest to  $N/n$ , we choose the units.

$\{U_{r+jk}\}$ , if  $r+jk \leq N$  and  $\{U_{r+jk-N}\}$ , if  $r+jk > N$ ;  $j = 0, 1, 2, \dots, (n-1)$ .

Taking the earlier example of selecting a systematic sample of size 14 from a population of 100 units ( $N = 100$ ,  $k = 7$  and  $n = 15$ ) all the samples can be made to have a size of 15 by adopting the CSS. A random start of 5 will lead to the selection of a sample of the 15 units 5, 12, 19, 26, 33, 40, 47, 54, 61, 68, 75, 82, 89, 96 and 3 ( $96 + 7 - 100$ ). This procedure ensures equal probability of selection to every unit in the population.

*Besides constancy of the sample size from sample to sample, the CSS procedure ensures that  $m_r$  the sample mean is an unbiased estimate of the population mean.* (2.55)

Let  $nk = N$ . Then  $m^* = m$ . There are  $k$  possible samples, each sample with a probability of  $1/k$ . Let the sample mean of the  $r$ -th systematic sample be  $m_r = (1/n) \sum_{i=1}^n y_{ir}$ , where  $y_{ir}$  is the value of the characteristic under study for the  $i$ -th unit in the  $r$ -th systematic sample, summation is from  $i = 1$  to  $n$ . As already noted  $m_i$  is an unbiased estimator of  $M$  or  $E(m_r) = M$ . We thus have  $k$  possible unbiased estimates of  $M$ . Denoting the sample mean in systematic sampling as  $m_{sys}$ , the sampling variance of  $m_{sys}$ , and related results of interest are:

$$V(m_{sys}) = \sigma_b^2 \quad (\text{the between-sample variance}). \quad (2.56)$$

$$V(m_{sys}) = V(y) - \sigma_w^2, \quad \text{where } \sigma_w^2 \text{ is within-sample variance.} \quad (2.57)$$

Equation 2.57 shows that (i)  $V(m_{sys})$  is less than the variance of the variable under study or the population variance, since  $\sigma_w^2$  is  $> 0$  and (ii)  $V(m_{sys})$  can be reduced by increasing  $\sigma_w^2$ , or by increasing the within-sample variance. (ii) would happen *if the units within each systematic sample are as heterogeneous as*

*possible*. Since we select a sample of 'n' units from the population of N units by selecting every k-th element from the random start 'r', the population is divided into 'n' groups and we select one unit from each of these 'n' groups of population units. Units *within* a sample would be heterogeneous if there is heterogeneity *between* the 'n' groups. This would imply that units *within* each of the n groups would have to be as homogeneous as possible. All these suggest that the sampling variance of the sample mean is related to the *arrangement of the units in the population*. This is both an advantage and disadvantage of systematic sampling. An arrangement that conforms to the conditions mentioned above would lead to a smaller sampling variance or an efficient estimate of the population mean while a 'bad' arrangement would lead estimates that are not as efficient.

#### 7.5.4.5 Sampling with Probability Proportional to Size (PPS)

**The Sampling Method:** We have so far considered sampling methods in which the probability of each unit in the population getting selected in the sample was equal. There are also methods of sampling in which the probability of any unit in the population getting included in the sample varies from unit to unit. One such method is sampling with probability proportional to size (pps) in which the probability of selection of a unit is proportional to a given measure of its size. This measure may be a characteristic related to the variable under study. One example may be the employment size of a factory in the past year and the variable under study may be the current year's output. Does this method lead to a bias in our results, as units with smaller sizes would be under represented in the sample and those with larger sizes would be over represented. It is true that if the sample mean 'm' were to be used to estimate the population mean M, m would be a biased estimator of M. However, *what is done in this method of sampling is to weight the sample observations with suitable weights at the estimation stage to obtain unbiased estimates of population parameters, the weights being the probabilities of selection of the units.*

**Estimates from pps sample of size 1:** Let the population units be  $\{U_1, U_2, \dots, U_N\}$ . Let the main variable Y and the related size variable X associated with these units be  $\{Y_1, X_1; Y_2, X_2; \dots, Y_N, X_N\}$ . The probability of selecting any unit, say,  $U_i$  in the sample will be  $P_i = (X_i / X)$ , where  $\sum_{i=1}^N X_i = X$ , where  $i = 1$  to  $N$ . Let us select *one unit* by pps method. Let the unit selected thus have the values  $y_1$  and  $x_1$  for the variables y and x. The variables y and x are random variables assuming values  $Y_i$  and  $X_i$  respectively with probabilities  $P_i$ ,  $i = 1, 2, \dots, N$ . The following results based on the sample of size 1 can be derived easily:

$$\text{An unbiased estimator of population total Y is } Y^*_{(1)pps} = y_1 / p_1 \quad (2.58)$$

$$\text{An unbiased estimator of M is } m^*_{(1)pps} = (1/N) Y^*_{(1)pps} = (1/N)(y_1 / p_1) \quad (2.59)$$

$$V[Y^*_{(1)pps}] = \sum_{i=1}^N (Y_i^2 / P_i) - Y^2 \quad (2.60)$$

$$V[m^*_{(1)pps}] = (1/N^2) V[Y^*_{(1)pps}] = (1/N^2) [\sum_{i=1}^N (Y_i^2 / P_i) - Y^2] \quad (2.61)$$

These show that *the variance of the estimate will be small if the  $P_i$  are proportional to  $Y_i$ .*

**Estimators from pps sample of size > 1 [pps with replacement (pps-wr)]**